IIT Guwahati

# Advanced Certification Programme in Artificial Intelligence and Machine Learning

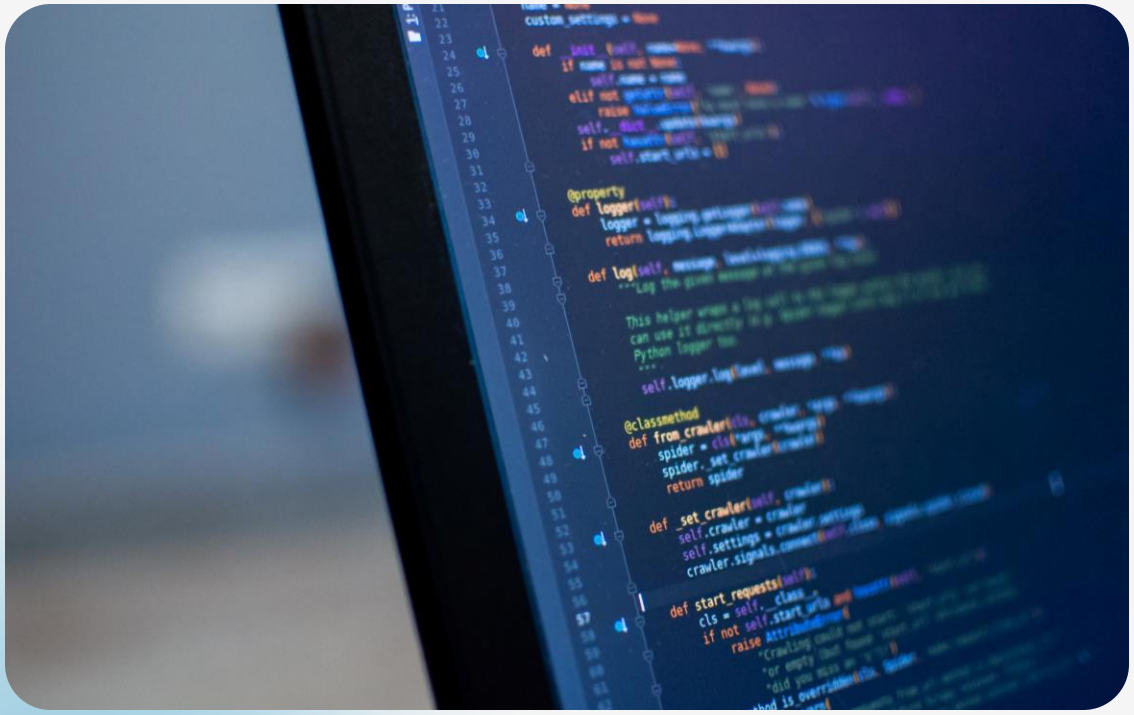
# Week 3: Python Fundamentals

# Topics Covered

- Python Fundamentals
- Mini Projects
- Case Study

# Python Fundamentals

# Control Structures in Python



**Key Functions of Control Structures:**

- Decision-Making: Implement conditional logic within programs
- Flow Control: Direct the order of statement execution
- Application Responsiveness: Build dynamic and interactive applications

# Understanding if-else Statements

```
age = 18
if age >= 18:
    print("You are an adult.")
else:
    print("You are not an adult.")
```

if-else statements execute different code blocks based on whether a condition is true or false.

How if-else works:

- **Conditional Check:** Evaluates a specified condition.
- **True Execution:** Runs code within the if block if the condition is true.
- **False Execution:** Runs code within the else block if the condition is false.

# Using if-elif-else for Multiple Conditions

```
score = 85
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
else:
    print("Grade: D")
```

**Key Features of if-elif-else:**

- **Sequential Evaluation:** Conditions are checked in order
- **Exclusive Execution:** Only one code block executes (the first true condition)
- **Default Behaviour:** The else block executes if no other condition is met

# Nested if Statements

```
num = 10
if num > 0:
    print("Positive number")
    if num % 2 == 0:
        print("Even number")
    else:
        print("Odd number")
else:
    print("Negative number")
```

Nested if statements embed if statements within other if or else blocks, creating hierarchical decision structures.

**Purpose of Nested if Statements:**

- **Implement Complex Logic:** Handle scenarios with multiple dependent conditions
- **Create Hierarchical Decisions:** Structure decision-making into levels of increasing specificity
- **Refine Conditional Checks:** Perform further checks based on the outcome of previous conditions

# Practical Applications of Control Structures


**Real-World Uses:**

- **Form Validation**: Ensure accurate user input
- **Game Development**: Implement game logic and respond to player actions
- **Data Processing**: Perform different operations based on data characteristics
- **Automation Scripts**: Automate tasks by executing specific actions based on predefined conditions

# Best Practices for Using Control Structures

## Writing Effective Control Structures


- Keep conditions concise and easy to understand
- Minimise nesting to enhance readability
- Employ functions or dictionary-based "switches" for improved clarity and organisation in complex scenarios
- Use descriptive variable names to make conditions self-explanatory

# Understanding File Handling in Python

## Opening a File

### File Access in Python

- Python's file handling enables interaction with computer files.
- It facilitates data storage and manipulation.
- The open() function is used for file handling.
- open() requires:
  - The filename.
  - The opening mode

### Common File Opening Modes

- **'r' (Read):** Opens the file for reading (default mode)
- **'w' (Write):** Opens the file for writing, creating a new file or overwriting an existing one
- **'a' (Append):** Opens the file for appending, adding new data to the end
- **'b' (Binary):** Opens the file in binary mode

# Understanding File Handling in Python

## Reading Files

### Code Example

```
file = open('example.txt', 'r')
content = file.read()  # Read entire file
print(content)
file.close()
```

### Key Reading Methods

- **read():** Reads the entire file content as a single string
- **readline():** Reads a single line from the file, including the newline character (\n)
- **readlines():** Reads all lines from the file and returns them as a list of strings

# Understanding File Handling in Python

## Writing and Appending to Files

### Key Writing Methods

```
file = open('example.txt', 'w')
file.write('Hello, world!\n')
file.write('This is a new line.')
file.close()
```

- **write():** Writes a single string to the file
- **writelines():** Writes a list of strings to the file

### The Append Process

```
file = open('example.txt', 'a')
file.write('\nAppending new content.')
file.close()
```

- **Open in Append Mode:** Use the 'a' mode with the open() function
- The new data to the file will be added after the existing content

# Using the with Statement for File Handling

## Code Example

```
with open('example.txt', 'r') as file:
    content = file.read()
    print(content)
# No need to call file.close()
```

## Benefits of Using with

- **Automatic File Closure:** Guarantees the file is closed, even if errors occur
- **Simplified Syntax:** Reduces boilerplate code, making file handling more concise
- **Improved Code Reliability:** Minimises the risk of file corruption or resource leaks.

# What are Comprehensions in Python?

## Key Concepts

- Comprehensions offer a concise and efficient way to create lists, dictionaries, and sets in Python
- It improves readability and performance compared to traditional loops

## Benefits of Comprehensions

- **Simplified Creation:** Streamline code for generating lists, dictionaries, and sets
- **Enhanced Readability:** Improve code clarity and expressiveness
- **Performance Efficiency:** Often provide faster execution than equivalent loops

# List Comprehension

## Code Example

```
# Traditional for loop approach
numbers = [1, 2, 3, 4, 5]
squares = []
for num in numbers:
    squares.append(num ** 2)
print(squares)  # Output: [1, 4, 9, 16, 25]

# List comprehension approach
squares = [num ** 2 for num in numbers]
print(squares)  # Output: [1, 4, 9, 16, 25]
```

## Features of List Comprehensions

- It provide a concise way to create new lists from existing iterables
- **Syntax:**
  [expression for item in iterable]
- **Optional Condition:**
  [expression for item in iterable if condition]
- **Improved Readability:** More readable than equivalent for loops

# Advanced List Comprehensions

## Code Example

```
# List comprehension with a condition
even_squares = [num ** 2 for num in
range(10) if num % 2 == 0]
print(even_squares)  # Output: [0, 4, 16, 36,
64]

# Nested list comprehension
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
flat = [num for row in matrix for num in
row]
print(flat)  # Output: [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

## Advanced Features

- List comprehensions can be enhanced with conditional filtering and nesting for more complex list creation
- **Conditional Filtering:** [expression for item in iterable if condition] filters elements based on a condition
- **Nested Comprehensions:** Create multi-dimensional data structures or flatten nested lists

# Dictionary Comprehension

## Code Example

```
# Traditional approach
numbers = [1, 2, 3, 4]
squared_dict = {}
for num in numbers:
    squared_dict[num] = num ** 2
print(squared_dict)  # Output: {1: 1, 2: 4, 3: 9, 4: 16}

# Dictionary comprehension approach
squared_dict = {num: num ** 2 for num in numbers}
print(squared_dict)  # Output: {1: 1, 2: 4, 3: 9, 4: 16}
```

## Dictionary Comprehensions Features

- It provides a concise way to create dictionaries from iterables by transforming keys and values
- **Basic Syntax:** {key_expression: value_expression for item in iterable}
- **Optional Condition:** {key_expression: value_expression for item in iterable if condition}

# Advanced Dictionary Comprehensions

## Code Example

```
# Dictionary comprehension with a condition
squared_even_dict = {num: num ** 2 for num in range(10) if num % 2 == 0}
print(squared_even_dict)  # Output: {0: 0, 2: 4, 4: 16, 6: 36, 8: 64}
```

## Advanced Features

- Dictionary comprehensions can include conditions to filter key-value pairs
- **Syntax with a condition:** {key_expression: value_expression for item in iterable if condition}

# Benefits of Using Comprehensions

Comprehensions offer several advantages over traditional loops when creating collections like lists, dictionaries, and sets.

- **Conciseness**: Achieve more with less code
- **Readability**: Enhance code clarity and understanding at a glance
- **Performance**: Often execute faster than equivalent loops and method calls
- **Expressiveness**: Combine filtering, mapping, and reducing operations in a single, readable line

# Best Practices for Using Comprehensions

## Writing Effective Comprehensions

- **Prioritise Readability:** Keep comprehensions simple and easy to understand
- **Avoid Over-Nesting:** If nesting significantly reduces readability, use standard loops instead
- **Use for Simple Operations:** Employ comprehensions for straightforward filtering and mapping tasks
- **Comment Complex Logic:** For more complex comprehensions, add comments to explain the logic

# Object-Oriented Programming (OOP)

## Key Principles

- Object-Oriented Programming (OOP) is a way of structuring code
- It organises code into reusable units called "objects“
- Each object combines data (its properties) and the actions it can perform (its methods)

## Features and Benefits

- **Organizes Code**: Structures code into objects that represent real-world entities.
- **Promotes Reusability**: Encourages code reuse and modularity.
- **Supports Scalability**: Facilitates easier maintenance and expansion of code.
- **Four Core Principles**: Encapsulation, Abstraction, Inheritance, and Polymorphism

# Understanding Classes and Objects

## Code Example

```
class Dog:
    def __init__(self, name, breed):
        self.name = name
        self.breed = breed

    def bark(self):
        return f"{self.name} says woof!"

my_dog = Dog('Buddy', 'Golden
Retriever')
print(my_dog.bark())  # Output: Buddy
says woof!
```

## Key Concepts

- **Class:** A template for creating objects (like a blueprint)
- **Object:** An instance of a class (a specific creation based on the blueprint)
- **Attributes:** Variables that store data specific to an object
- **Methods:** Functions that define the object's behaviours or actions

# The __init__ Method

## Constructor

### Code Example

```
class Car:
    def __init__(self, make, model):
        self.make = make
        self.model = model

    def display_info(self):
        return f"Car: {self.make} {self.model}"

car1 = Car('Toyota', 'Camry')
print(car1.display_info())  # Output: Car:
Toyota Camry
```

### Key Functions of __init__

- **Automatic Invocation:** Automatically called when a new object is created (instantiated)
- **Attribute Initialisation:** Assigns initial values to an object's attributes
- **Object Setup:** Sets up the initial state of an object

# Inheritance in Python

## Code Example

```
class Animal:
    def __init__(self, name):
        self.name = name

    def speak(self):
        pass

class Dog(Animal):
    def speak(self):
        return f"{self.name} says woof!"

my_pet = Dog('Max')
print(my_pet.speak())  # Output: Max says woof!
```

## Key Concepts

- **Base Class (Parent):** The class whose properties are inherited.
- **Derived Class (Child):** The class that inherits properties from the base class.
- **Extending/Modifying Behaviour:** Allows for extending or modifying the behaviour of inherited classes.

# Polymorphism in Python

## Code Example

```
class Bird:
    def speak(self):
        return "Tweet!"
class Cat:
    def speak(self):
        return "Meow!"
animals = [Bird(), Cat()]
for animal in animals:
    print(animal.speak())

# Output:
# Tweet!
# Meow!
```

## Key Aspects of Polymorphism

- **Unified Interface:** A single method name can be used across different classes
- **Varying Behaviour:** The method's implementation depends on the object's class
- **Achieved Through Overriding:** In Python, polymorphism is primarily achieved through method overriding.

# Encapsulation and Abstraction

## Code Example

```
class BankAccount:
    def __init__(self, balance):
        self.__balance = balance  # Private attribute

    def deposit(self, amount):
        if amount > 0:
            self.__balance += amount
        return self.__balance

account = BankAccount(1000)
print(account.deposit(500))  # Output: 1500
# print(account.__balance)  # Error:
AttributeError
```

## Key Aspects of Polymorphism

- **Encapsulation:** Bundles data (attributes) and methods that operate on that data within a single unit (a class), restricting direct access to some components

- **Abstraction:** Hides complex implementation details and exposes only essential information or interfaces to the user

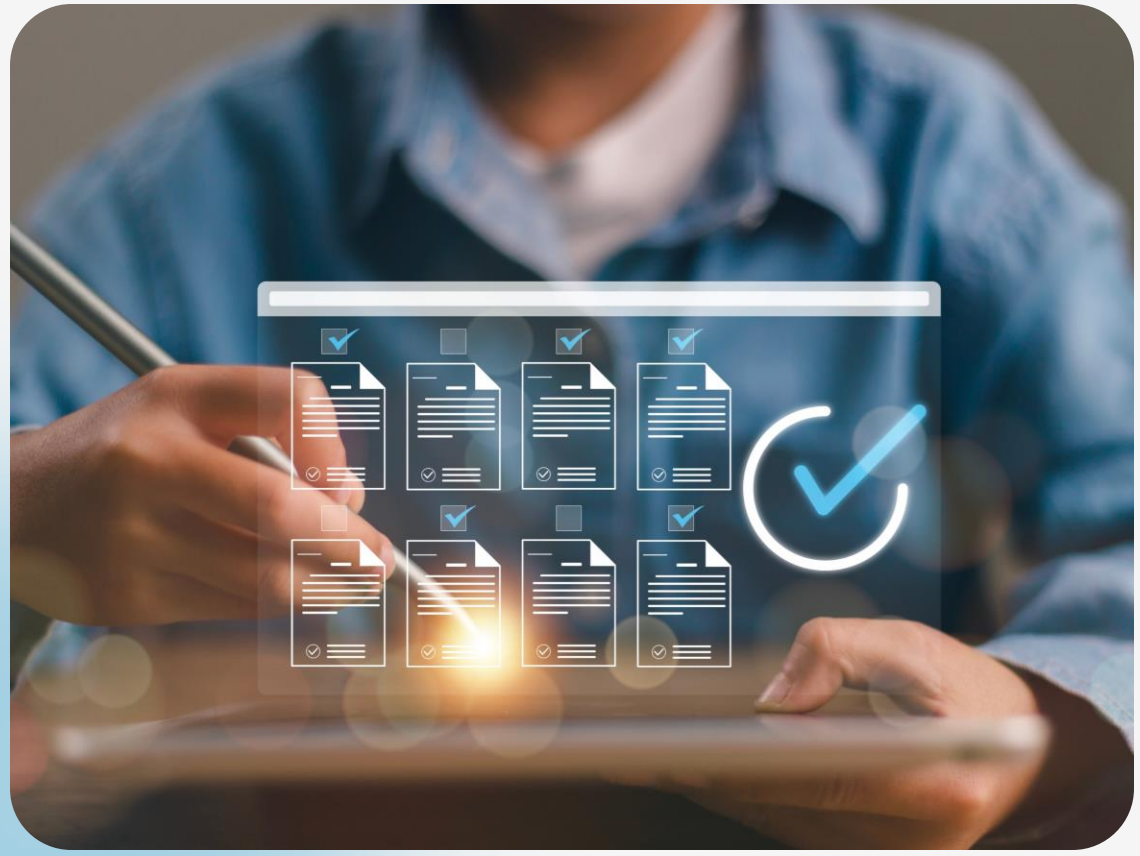
# Real-World Applications of OOP

OOP principles ensure cleaner, maintainable, and robust code for diverse applications.

- **Game Development:** Creating game entities with defined behaviours and interactions
- **Web Development:** Structuring web applications using models for users, data, and server-side logic
- **Data Science:** Organising data analysis and manipulation workflows using classes and objects
- **GUI Applications:** Building interactive user interfaces with reusable UI components

# Best Practices for OOP in Python

Adhering to these best practices will lead to cleaner, more maintainable, and robust object-oriented code.



- Use meaningful class and method names for better readability
- Single Responsibility Principle (SRP): Each class should have a single, well-defined purpose
- Avoid deep inheritance hierarchies; prefer composition over inheritance whenever possible
- Use docstrings to document classes and methods for better understanding

# Generators in Python

## Key Concepts

- Generators are a special type of iterable
- It produce a sequence of values one at a time, without storing the entire sequence in memory

## Characteristics of Generators

- **yield Keyword:** Use yield instead of return to produce values iteratively
- **Lazy Evaluation:** Values are generated only when requested (on demand)
- **Memory Efficiency:** Ideal for working with large datasets, as they use minimal memory

# Creating a Generator

Generators are created using functions, but instead of return, they use the yield keyword to produce a sequence of values iteratively.

## Code Example

```
def countdown(n):
    while n > 0:
        yield n
        n -= 1

for number in countdown(5):
    print(number)
# Output: 5, 4, 3, 2, 1
```

## Defining a Generator Function

- The *yield* keyword pauses execution and returns a value
- Each call to next() resumes execution from where it left off, yielding the next value
- Generators can be used directly in for loops.

# Differences Between Generators and Regular Functions

## Regular Functions:

- Use return to send a single value and then terminate.
- Start execution from the beginning each time they are called.
- Store all calculated values in memory (if creating a collection like a list).

## Generators:

- Use yield to produce a value and pause execution, retaining their internal state
- Resume execution from where they left off when the next value is requested.
- Generate values on demand (lazy evaluation), saving memory.

# Generator Expressions

Generator expressions offer a concise syntax for creating generators, similar to list comprehensions but using parentheses instead of square brackets.

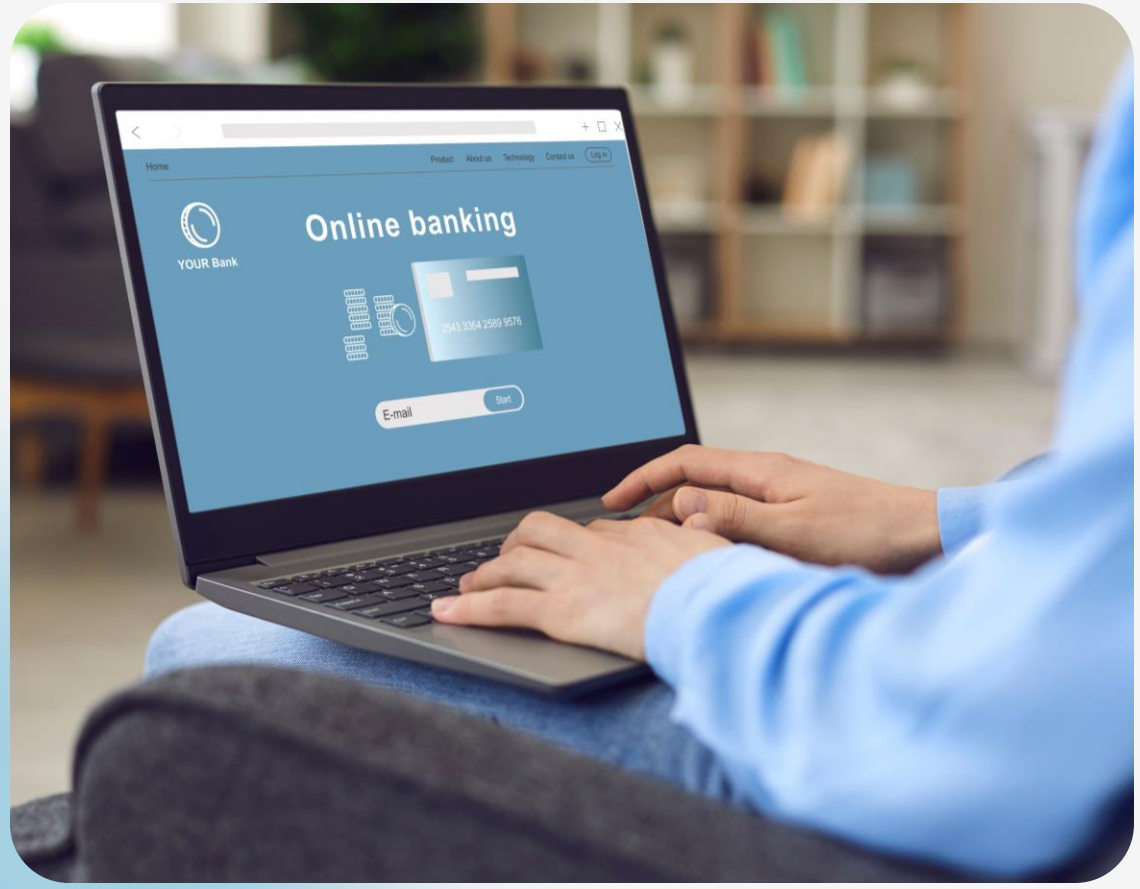
## Code Example

```
squares = (x * x for x in range(5))
for square in squares:
    print(square)
# Output: 0, 1, 4, 9, 16
```

## Key Features

- **Concise Syntax:** (expression for item in iterable)
- **Memory Efficiency:** Generates values on demand, similar to generator functions.
- **Lazy Evaluation:** Values are produced only when iterated over.

# Practical Applications of Generators




**Real-World Uses:**

**Large Datasets:** Process large files line by line without loading the entire file into memory

**Infinite Sequences:** Generate infinite sequences (e.g., Fibonacci numbers) without memory exhaustion

**Data Pipelines:** Create efficient data processing pipelines that generate intermediate results step by step

**Asynchronous Programming:** Use with async and await for efficient management of asynchronous tasks

# Best Practices for Using Generators

**Effective Generator Usages:**

**For Large Datasets:** Use generators to process large datasets and avoid memory overload

**itertools:** Combine generators with the itertools module for complex iteration patterns

**Generator Exhaustion:** Be aware that once a generator is exhausted, it cannot be reused

**Avoid Overuse:** For simple, non-memory-intensive tasks, standard loops might be more appropriate

# Why Python for Data Science?

## Key Features

Python has become the leading language for data science due to its:

- Simplicity
- Readability
- Powerful libraries

## Key Advantages

- **Extensive Community Support:** A large and active community provides ample resources and support
- **Powerful Libraries:** Specialized libraries (e.g., NumPy, Pandas, Scikit-learn) simplify complex data tasks
- **Seamless Integration:** Integrates well with other tools, languages, and technologies

# Introduction to NumPy

## Code Example

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print(arr * 2)

# Output: [ 2  4  6  8 10]
```

## Key Features of NumPy

- N-dimensional Arrays (ndarray) for efficient storage and manipulation of numerical data in array format
- Offers a rich set of mathematical functions, including linear algebra, random number generation, and more
- Integrates seamlessly with C/C++ and Fortran code for optimized performance

# Key Features of NumPy

## Code Example

```
matrix = np.array([[1, 2], [3, 4]])
print(np.linalg.inv(matrix))  # Inverse of the matrix
print(np.mean(matrix))  # Mean of all elements
```

## Core Functionalities

- **Array Manipulation:** Element-wise operations, broadcasting, slicing, and indexing
- **Math Functions:** Trigonometric, statistical, and linear algebra operations
- **Random Numbers:** Generation of random numbers for simulations and testing

# Introduction to Pandas

Pandas is a Python library for efficient data manipulation and analysis with structured data.

## Code Example

```
import pandas as pd
data = {'Name': ['John', 'Anna', 'Peter'],
'Age': [28, 24, 35]}
df = pd.DataFrame(data)
print(df)

# Output:
#    Name  Age
# 0  John   28
# 1  Anna   24
# 2  Peter  35
```

## Key Features of Pandas

- **DataFrame:** A 2D labeled data structure, similar to a table in SQL or Excel
- **Series:** A 1D labeled array, analogous to a column in a table
- **Data Handling:** Supports data alignment, missing data handling, and group operations

# Key Features of Pandas

## Code Example

```
# Filtering data
filtered_df = df[df['Age'] > 25]

# Grouping data
grouped = df.groupby('Age').mean()
```

## Core Features

- **Data Manipulation:** Filtering, sorting, and aggregating data
- **Missing Data Handling:** Detecting, filling, and dropping missing values
- **GroupBy Operations:** Splitting data into groups and applying aggregate functions

# Introduction to Matplotlib

Matplotlib is a Python library for static, animated, and interactive visualisations.

## Code Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [10, 20, 25, 30]

plt.plot(x, y)
plt.title('Simple Line Plot')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.show()
```

## Key Features

- **Diverse Plotting Options:** Supports a wide range of plot types
- **Highly Customisable:** Offers extensive control over plot appearance
- **Complex Layouts:** Enables the creation of subplots and complex figure layouts

# Key Features of Matplotlib

## Code Example

```
# Creating subplots
fig, axs = plt.subplots(2, 2)
axs[0, 0].plot(x, y)
axs[0, 1].scatter(x, y)
axs[1, 0].bar(x, y)
axs[1, 1].hist(y)
plt.show()
```

## Core Capabilities

- **Diverse Plot Types:** Supports various plot types
- **Extensive Customisation:** Fine-grained control over plot appearance
- **Subplots and Layouts:** Enables the creation of complex figures with multiple subplots and custom layouts

# Mini Projects

# Mini Project 1: Library Management System

## Project Overview

- Create a simple library management system
- Allow users to add, remove, search for, and borrow books
- Maintain a record of borrowed books and their borrowers

# Action Items for Group Members: Member 1

## Data Structures and Basic Functionality

| Feature | Action |
| --- | --- |
| Lists and Tuples | Use a list of tuples to store book information (title, author, ISBN) |
| Basic Functions | Write functions for adding, removing, and listing books |
| Sets and Dictionaries | • Use a set to prevent duplicate books<br>• Use a dictionary to map borrowed books to borrowers |
| Collaboration | Collaborate with Member 2 on data structure and duplicate handling |

# Action Items for Group Members: Member 2

## Advanced Functionalities and File Handling

| Feature | Action |
| --- | --- |
| File Handling | • Read and write to a file in CSV format for persistent storage<br>• Load data from the file on startup and save changes on exit |
| Switch Cases | Use a dictionary-based approach for menu selection handling |
| Control Structures and Loops | • Create a menu-driven interface using loops and if/else statements<br>• Allow users to add, remove, and borrow books |
| Collaboration | Collaborate with Member 1 on integrating file handling |

# Action Items for Group Members: Member 3

## OOP and Generators

| Feature | Action |
| --- | --- |
| OOP Concepts | • Create a Book class with attributes like title, author, ISBN, and status<br>• Create a Library class with methods for managing books |
| List and Dictionary Comprehensions | • Filter books based on criteria (e.g., author, genre)<br>• Create a summary report of borrowed books |
| Generators | Implement a generator to iterate over available books |
| Collaboration | Collaborate with Members 1 and 2 for integration |

# Mini Project 2: Personal Expense Tracker

## Project Overview

- Develop a personal expense tracker
- Allow users to log expenses, categorize them, view summaries, and generate reports

# Action Items for Group Members: Member 1

## Data Structures and Basic Functionality

| Feature | Action |
| --- | --- |
| Lists and Tuples | Use a list of tuples to store expense details (date, amount, category, description) |
| Basic Functions | Write functions to add, delete, and list expenses |
| Sets and Dictionaries | • Use a set for unique categories<br>• Use a dictionary to map categories to expenses |
| Collaboration | Collaborate with Member 2 on data structures and functions |

# Action Items for Group Members: Member 2

## User Interface and File Management

| Feature | Action |
| --- | --- |
| File Handling | • Save and load expenses using CSV or JSON format<br>• Ensure data persistence between sessions |
| Switch Cases | • Use a dictionary-based approach for menu handling |
| Control Structures and Loops | Create a command-line interface for navigating through options |
| Collaboration | Collaborate with Member 1 to integrate file handling |

# Action Items for Group Members: Member 3

## Advanced Features and Reporting

| Feature | Action |
| --- | --- |
| OOP Concepts | Create Expense and ExpenseTracker classes |
| List and Dictionary Comprehensions | • Filter expenses based on criteria<br>• Generate summaries (e.g., total expenses per category) |
| Generators | Implement a generator for efficient report generation |
| Collaboration | Collaborate with Members 1 and 2 for seamless integration |

# Mini Project 3: Simple Quiz Application

## Project Overview

- Create a quiz application where users take quizzes on various topics
- Record scores and display performance

# Action Items for Group Members: Member 1

## Data Management and Quiz Setup

| Feature | Action |
| --- | --- |
| Lists and Tuples | Store quiz questions with options and correct answers in tuples |
| Basic Functions | Add, delete, and retrieve questions by topic |
| Sets and Dictionaries | • Use a set for quiz topics/categories<br>• Use a dictionary to map topics to their questions |
| Collaboration | Collaborate with Member 2 on quiz management |

# Action Items for Group Members: Member 2

## User Interface and Scoring System

| Feature | Action |
| --- | --- |
| File Handling | • Save and load user scores to/from a file<br>• Ensure data persistence between sessions |
| Switch Cases | Use a dictionary-based approach for handling quiz options |
| Control Structures and Loops | Create a command-line interface for taking quizzes |
| Collaboration | Collaborate with Member 1 to integrate file handling |

# Action Items for Group Members: Member 3

## Advanced Quiz Features and Reporting

| Feature | Action |
| --- | --- |
| OOP Concepts | Create Question and Quiz classes |
| List and Dictionary Comprehensions | Filter questions and generate performance reports |
| Generators | Implement a generator for efficient question iteration |
| Collaboration | Collaborate with Members 1 and 2 for seamless integration |

# Case Study

# Introduction to Case Study

## Objective of the Report

Illustrate a real-world example where Python was utilized to solve a problem in an industry

Demonstrate how Python optimised a process or innovated a product/service

Provide a comprehensive overview including context, challenges faced, Python tools and techniques used, and results achieved

# Case Study Report Outline

**Introduction**

- Industry & company overview
- Context: problem/need for python
- Significance of the case study

**Background and Problem Statement**

- Problem/challenge faced
- Criticality of addressing the problem
- Previous solutions & limitations

**Role of Python**

- Python as the solution tool
- Libraries/frameworks used
- Python-based solution
- Key features

# Q & A

# Thank You